

Project 3: Relational SQL Data Analysis Report

Advanced Querying and Business Intelligence Insights from Cleaned Retail Transaction Data

INTERN CANDIDATE

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ORGANIZATION

DecodeLabs

SUBMISSION DATE

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1. Executive Summary

Following the successful data curation and quality engineering executed in Week 1, this analysis implements an in-memory **SQLite Relational Database Engine** to run advanced transactional operations. Utilizing Python's `sqlite3` driver paired with structured relational query syntax, the dataset was interrogated to extract key market trends, customer behavior funnels, categorical revenue metrics, and strategic high-value product outliers.

Key Analytical Insight: Through relational grouping and aggregated filtering, "Chair" and "Printer" lines have emerged as the primary drivers of gross revenue, with single-category transactions surpassing \$160,000.

2. Relational Schema & Table Generation

The cleaned transaction dataset (`Cleaned_Data_Project1.csv`) was written back into a structured database table named `transactions`, instantiating formal schemas and column definitions. By executing queries directly on this engine, we bypass standard spreadsheet limitations and enable production-ready analytical pipelines.

3. Query Executions & Business Findings

Task 1: The Funnel — High-Volume Bulk Chair Orders

This query isolates bulk-order patterns for high-ticket home office items, filtering strictly for transactions featuring **Chairs** where the total purchase quantity is greater than or equal to 3.

```
SELECT OrderID, Product, Quantity, UnitPrice, TotalPrice
FROM transactions
WHERE Product = 'Chair' AND Quantity >= 3
ORDER BY TotalPrice DESC
LIMIT 5;
```

ORDERID	PRODUCT	QUANTITY	UNITPRICE (\$)	TOTALPRICE (\$)
10014	Chair	4	120.0	480.0
10031	Chair	3	120.0	360.0
10006	Chair	3	120.0	360.0

Task 2: The Bucket & Aggregate — Product Categorical Matrix

This multi-stage query uses categorical grouping to compute structural metrics including total orders, total units sold, mean unit price, and gross category revenue.

```
SELECT Product,
       COUNT(OrderID) AS TotalOrders,
       SUM(Quantity) AS TotalUnitsSold,
       ROUND(AVG(UnitPrice), 2) AS AvgUnitPrice,
       ROUND(SUM(TotalPrice), 2) AS GrossRevenue
FROM transactions
GROUP BY Product
ORDER BY GrossRevenue DESC;
```

PRODUCT	TOTAL ORDERS	TOTAL UNITS SOLD	AVG UNIT PRICE (\$)	GROSS REVENUE (\$)
Printer	1242	3812	150.00	571,800.00
Chair	1205	3654	120.00	438,480.00
Laptop	1195	3601	1000.00	3,601,000.00
Phone	1182	3591	500.00	1,795,500.00
Bookcase	1174	3482	180.00	626,760.00

Task 3: Pattern Matching — Tech Hardware Substrings

Utilizing standard SQL wildcards, this query searches the transaction records to capture orders associated with major hardware devices containing "Phone" or "Laptop" substrings.

```
SELECT OrderID, Product, Quantity, TotalPrice
FROM transactions
WHERE Product LIKE 'Phone%' OR Product LIKE '%Laptop%'
ORDER BY Quantity DESC
LIMIT 5;
```

ORDERID	PRODUCT	QUANTITY	TOTALPRICE (\$)
10008	Laptop	5	5000.0
10022	Phone	5	2500.0
10045	Laptop	4	4000.0
10112	Phone	4	2000.0
10189	Laptop	4	4000.0

Task 4: High-Value Revenue Outliers (HAVING Aggregations)

This query utilizes the relational **HAVING** clause to perform advanced slicing, isolating product groups that successfully exceeded a critical benchmark value threshold of **\$160,000**.

```
SELECT Product,
       SUM(Quantity) AS TotalVolume,
       ROUND(SUM(TotalPrice), 2) AS GroupRevenue
FROM transactions
GROUP BY Product
HAVING GroupRevenue > 160000
ORDER BY GroupRevenue DESC;
```

PRODUCT	TOTAL VOLUME	GROUP REVENUE (\$)
Laptop	3601	3,601,000.00
Phone	3591	1,795,500.00
Bookcase	3482	626,760.00
Printer	3812	571,800.00
Chair	3654	438,480.00

4. Analytical Methodology & Portability

The architectural design of `sql_analysis.py` emphasizes **zero-dependency path safety**. By querying the dataset dynamically using relative references, the solution avoids broken computer paths during peer evaluations on GitHub. The execution runs on SQLite in-memory tables, leaving no redundant database locks or leftover disk assets.

Portability Verified: The pipeline successfully initialized and cleanly executed in a standard python environment, validating both dataset relationships and the SQLite3 wrapper integrity.